

N1.

$$X \sim \text{Bin}(n, p)$$

$$M[X] = np$$

$$D[X] = np(1-p)$$

$$\frac{K_1 + K_2}{2} = M[X] \Rightarrow \text{отрезок } [K_1, K_2] \text{ симметричен относительно } M[X]$$

$$\alpha = K_2 - M[X]$$

$$M[X] - K_1 = \alpha, \quad K_2 - M[X] = \alpha$$

$$K_1 = M[X] - \alpha, \quad K_2 = M[X] + \alpha$$

$$K_1 \leq X \leq K_2 \Leftrightarrow M[X] - \alpha \leq X \leq M[X] + \alpha \Leftrightarrow |X - M[X]| \leq \alpha$$

$$P(K_1 \leq X \leq K_2) = P(|X - M[X]| \leq \alpha)$$

Неравенство Чебышёва:

$$P(|Y - M[Y]| \geq \varepsilon) \leq \frac{D[Y]}{\varepsilon^2}, \quad \varepsilon > 0$$

$\Downarrow$

$$P(|Y - M[Y]| < \varepsilon) \geq 1 - \frac{D[Y]}{\varepsilon^2}$$

$\Downarrow$

$$P(|X - M[X]| \leq \alpha) \geq 1 - \frac{D[X]}{\alpha^2}$$

$\Downarrow$

$$P(|X - M[X]| \leq \alpha) \geq 1 - \frac{np(1-p)}{(K_2 - M[X])^2}$$

$\Downarrow$

$$P(K_1 \leq X \leq K_2) \geq 1 - \frac{np(1-p)}{(K_2 - M[X])^2}$$

N2.

Пусть X_i - результат i -го броска

$$S_n = X_1 + \dots + X_n$$

$$P(S_n > 780) \geq 0,75, \quad n = ?$$

$$P(X_i = k) = \frac{1}{6}$$

$$M[X_i] = \frac{1+2+3+4+5+6}{6} = 3,5$$

$$M[X_i^2] = \frac{1+4+9+16+25+36}{6} = \frac{91}{6}$$

$$D[X_i] = M[X_i^2] - (M[X_i])^2 = \frac{91}{6} - \frac{49}{4} = \frac{182-147}{12} = \frac{35}{12}$$

$$\sigma = \sqrt{D[X_i]} = \sqrt{\frac{35}{12}}$$

$$M[S_n] = n \cdot M[X_i] = \frac{7}{2}n$$

$$D[S_n] = n \cdot D[X_i] = \frac{35}{12}n$$

$$\sigma_{S_n} = \sqrt{D[S_n]} = \sigma\sqrt{n} = \sqrt{\frac{35}{12}} \cdot \sqrt{n}$$

$$Z = \frac{S_n - M[S_n]}{\sigma_{S_n}} = \frac{S_n - \frac{7}{2}n}{\sigma\sqrt{n}}, \quad Z \sim N(0, 1)$$

$$P(S_n > 780) = P\left(\frac{S_n - \frac{7}{2}n}{\sigma\sqrt{n}} > \frac{780 - \frac{7}{2}n}{\sigma\sqrt{n}}\right) = P\left(Z > \frac{780 - \frac{7}{2}n}{\sigma\sqrt{n}}\right)$$

$$P\left(Z > \frac{780 - \frac{7}{2}n}{\sigma\sqrt{n}}\right) \geq 0,75; \quad P\left(Z > \frac{780 - \frac{7}{2}n}{\sigma\sqrt{n}}\right) = 1 - \Phi\left(\frac{780 - \frac{7}{2}n}{\sigma\sqrt{n}}\right)$$

$$1 - \Phi\left(\frac{780 - \frac{7}{2}n}{\sigma\sqrt{n}}\right) \geq 0,75 \Rightarrow \Phi\left(\frac{780 - \frac{7}{2}n}{\sigma\sqrt{n}}\right) \leq 0,25 \Rightarrow$$

$$\Rightarrow \frac{780 - \frac{7}{2}n}{\sigma\sqrt{n}} \approx -0,67$$

$$780 - \frac{7}{2}n = -0,67 \cdot \sqrt{\frac{35}{12}} \cdot \sqrt{n} \Leftrightarrow 780 - 3,5n + 1,144\sqrt{n} = 0$$

$$t = \sqrt{n}, \quad t \geq 0; \quad 3,5t^2 - 1,144t - 780 = 0$$

$$t_1 \approx -14,766$$

$$t_2 \approx 15,093.$$

$$\sqrt{n} = 15,093 \Rightarrow n \approx 227,8 \Rightarrow n \approx 228$$

Answer: 228.